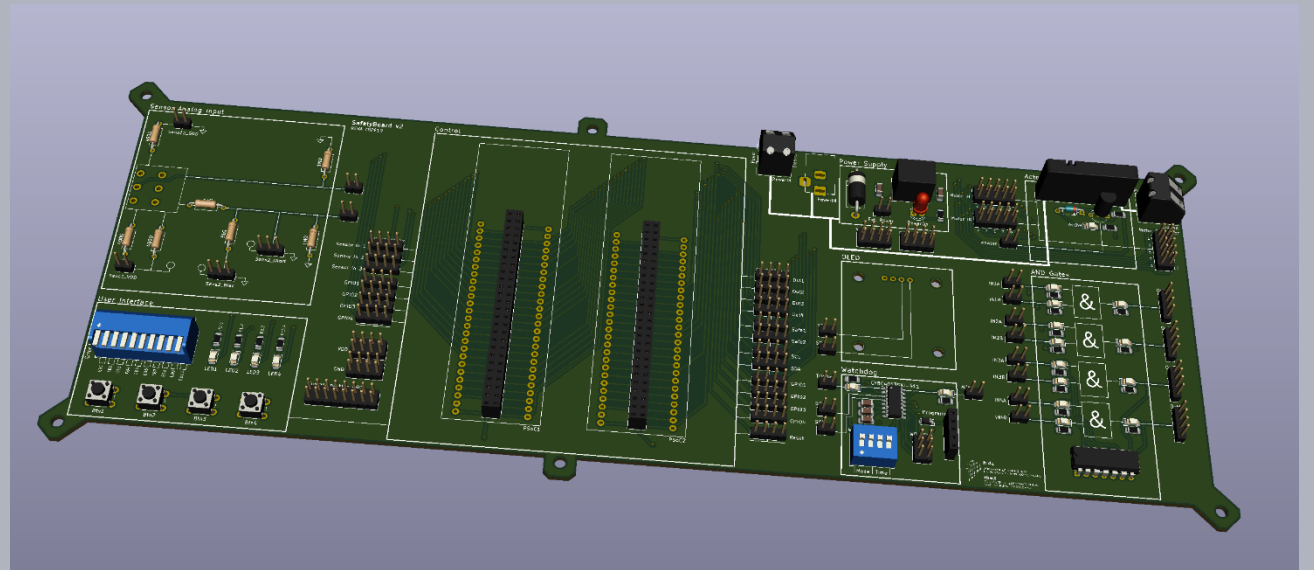


Safety in Embedded Control Systems

Standards

Prof. Dr.-Ing. Peter Fromm



Content

- Why do we need Standards
- A brief history of Safety Standards
- IEC 61508 Part 1 (Overall Safety Lifecycle)
 - Structure
 - Terms
 - Concepts
- Comparison IEC61508 and ISO26262

INTERNATIONAL
STANDARD

ISO
26262-6

First edition
2011-11-15

**Road vehicles — Functional safety —
Part 6:
Product development at the software
level**

Véhicules routiers — Sécurité fonctionnelle —

Partie 6: Développement du produit au niveau du logiciel

Last Lecture - Definition of Safety and Functional Safety

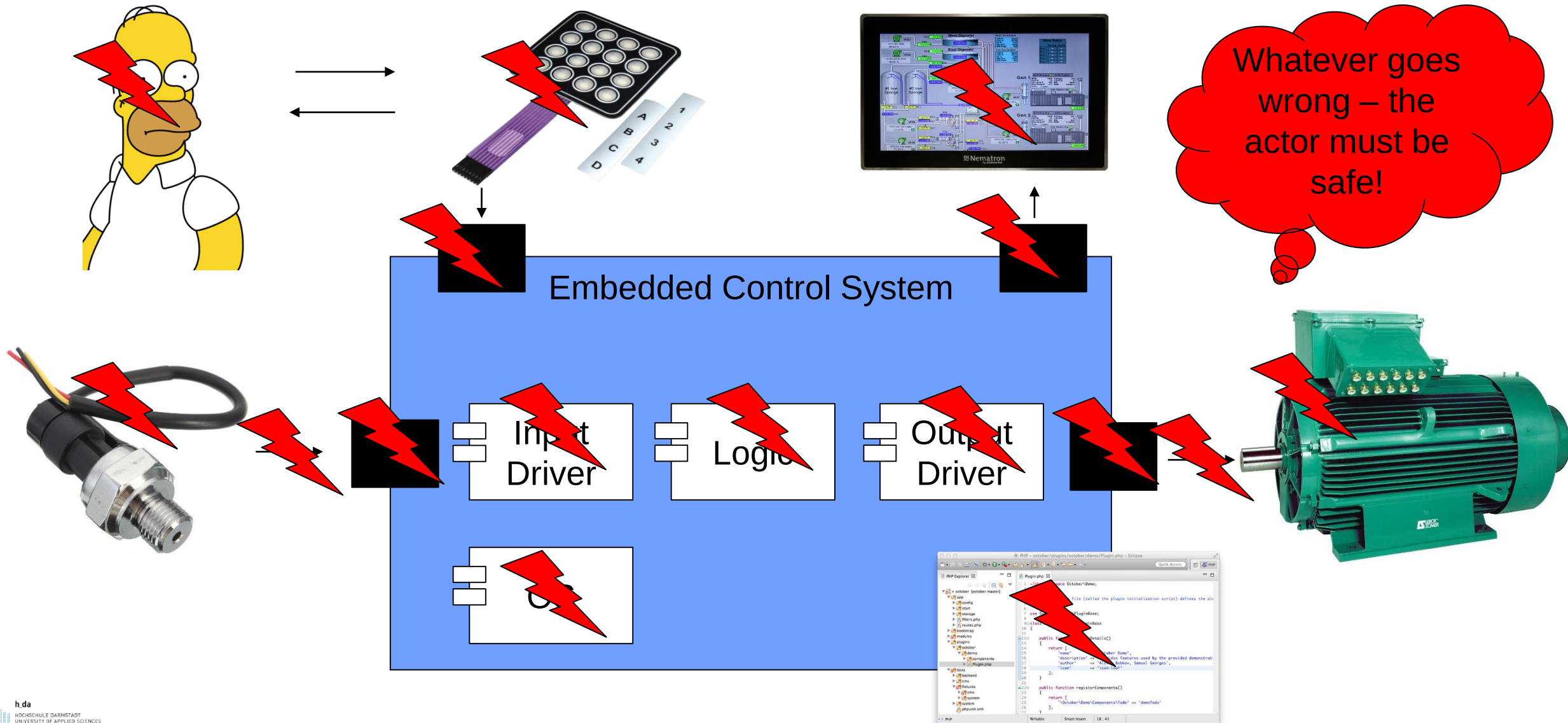
IEC 61508

- **Safety** is the freedom from **unacceptable risk** of physical **injury** or of damage to the **health of people**, either directly, or indirectly as a result of damage to property or to the environment.
- **Functional Safety** is part of the overall safety that depends on a system or equipment operating correctly (i.e. perform a **safety function**) in response to its inputs.

ISO26262

- functional safety: absence of **unreasonable risk** (1.136) due to **hazards** (1.57) caused by **malfunctioning behavior** (1.73) of **E/E systems** (1.31)

Last Lecture - Safety in Embedded Control Systems



Last Lecture - Safety function categories

Let's compare the 3 functions Airbag, Electronic Steering, Electronic Brake

	Airbag	Electronic Steering	Electronic Brake
Potential hazard	1) Does not explode during accident 2) Explodes during normal driving	1) Driving does not respond to steering wheel	1) Brake does not react 2) Brakes by itself
Call frequency	Only during accident	Permanently	Regularly
Safe state	Off (Why?)	None	Good question

Protection
function

Aims at reducing the risk in a dangerous situation

Safe system
function

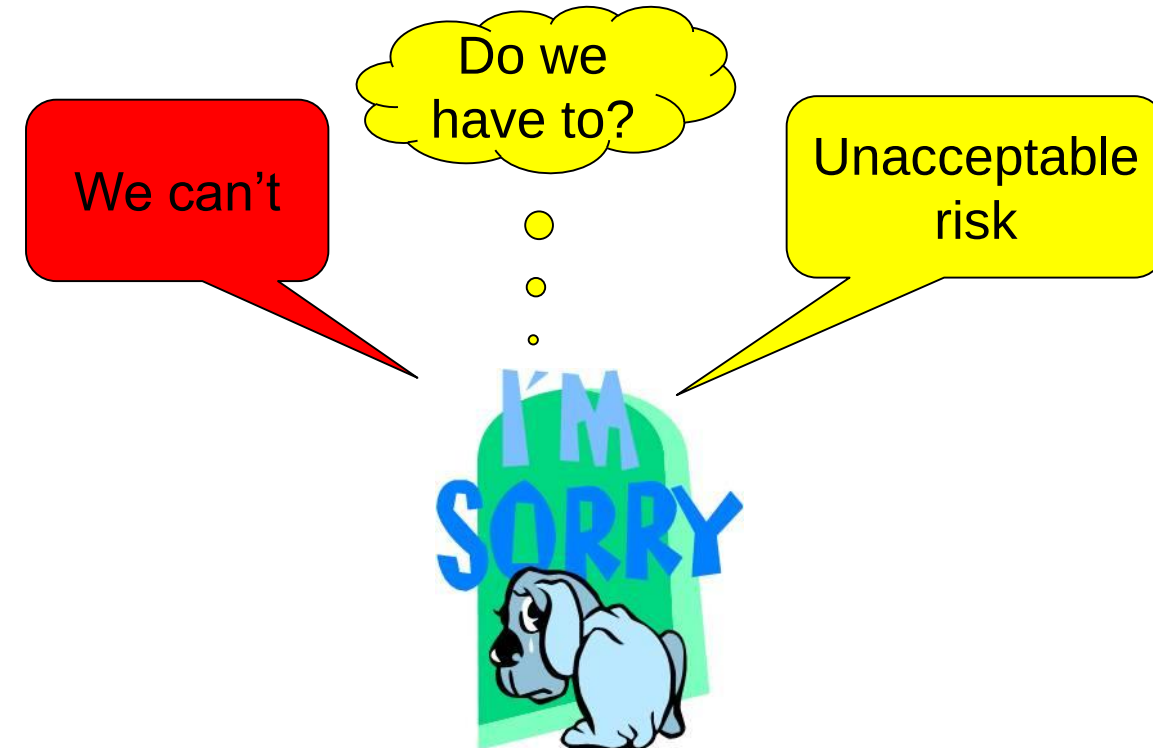
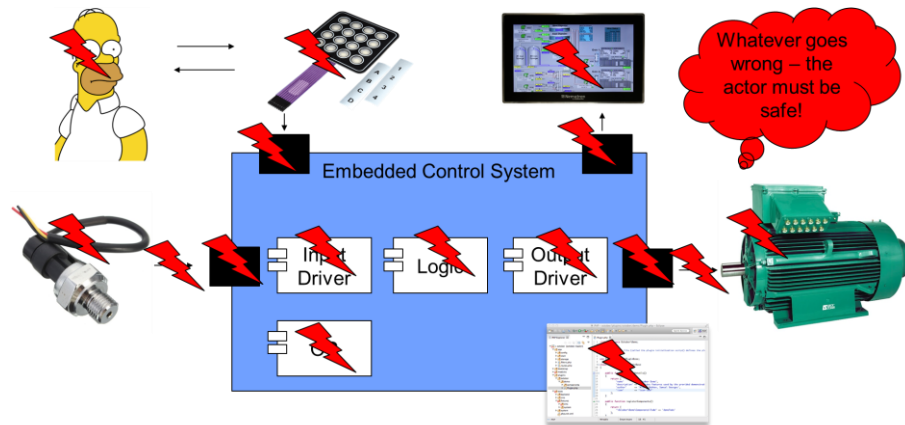
Must be safe during normal operation.

Safe system
function

Must be safe during normal operation.

Development of a safety critical system

- Can we develop a (complex) safety critical system which has no (ZERO) risk?
- Millions of transistors and lines of code?
- Operating under harsh conditions?
- Untrained user?
- Suffering from regular firmware updates?



What means “avoiding unacceptable risk”?

So, what can we do?

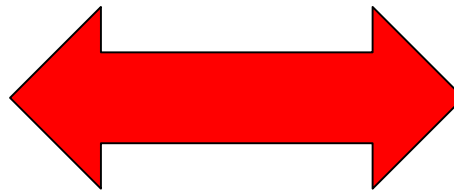
- Avoid systematic errors during development
- Supervision of the system to detect sporadic faults
- Safe handling of errors and transition into a safe state (if such a state exists)



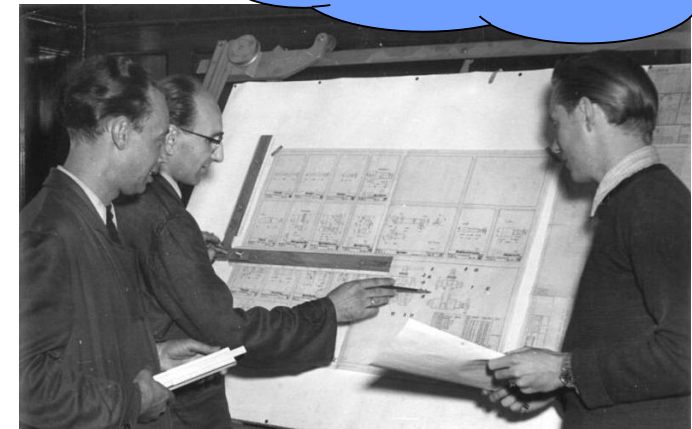
Fundamental Engineering Rule!

- Problem: In case of a real problem (people got killed or injured) a lawyer will have to judge.
 - Even bigger problem: A lawyer is not an engineer
- As long as we develop a system according the current technical state of the art, we are on the safe side.

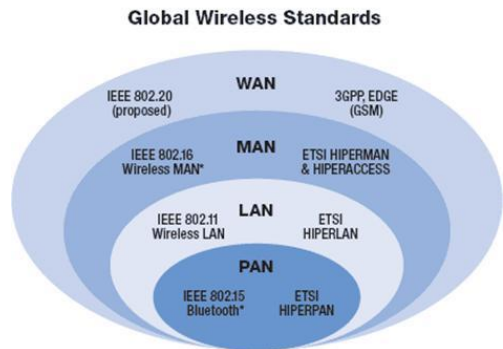
Laws (civil,
criminal, traffic...)



Standards



Why do we need standards?



Where do standards help – and where not

- Represents the official technical state of the art.
- Contain best practice recommendations (or obligations)
- Put a focus on processes and generic methods

- Might not be up to date with technologies.
- Do not provide ready-to-use solutions
- Specific recommendations and architectural concepts often are less prominent
- Are hard to read and often require additional training

As an engineer, you take decisions. Don't blame the standard if your system does not work!

Which Standard should we take?

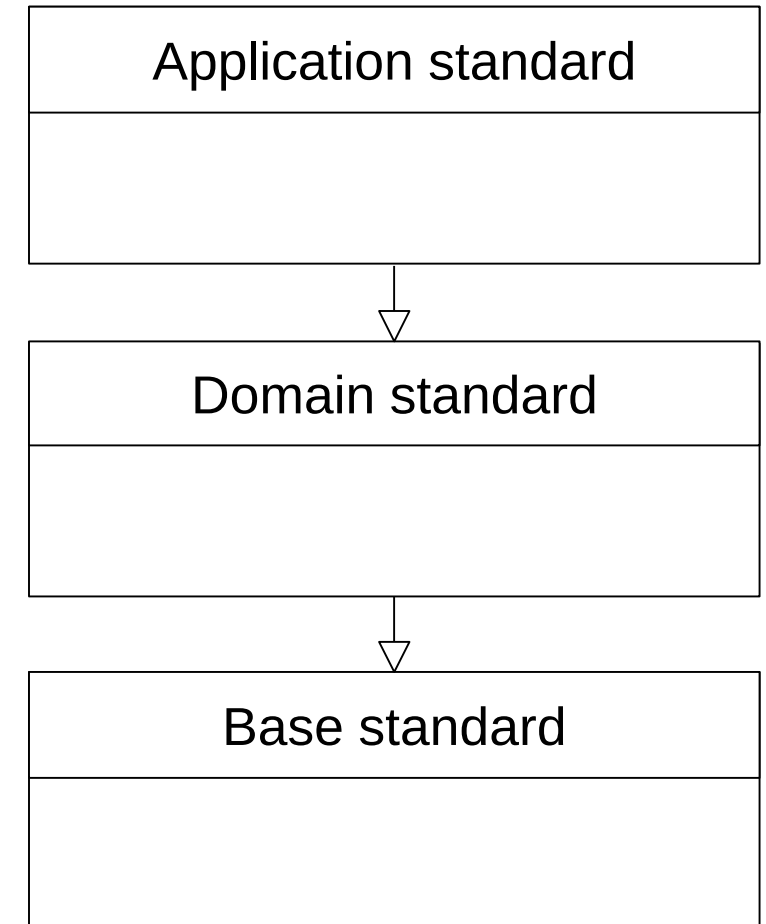
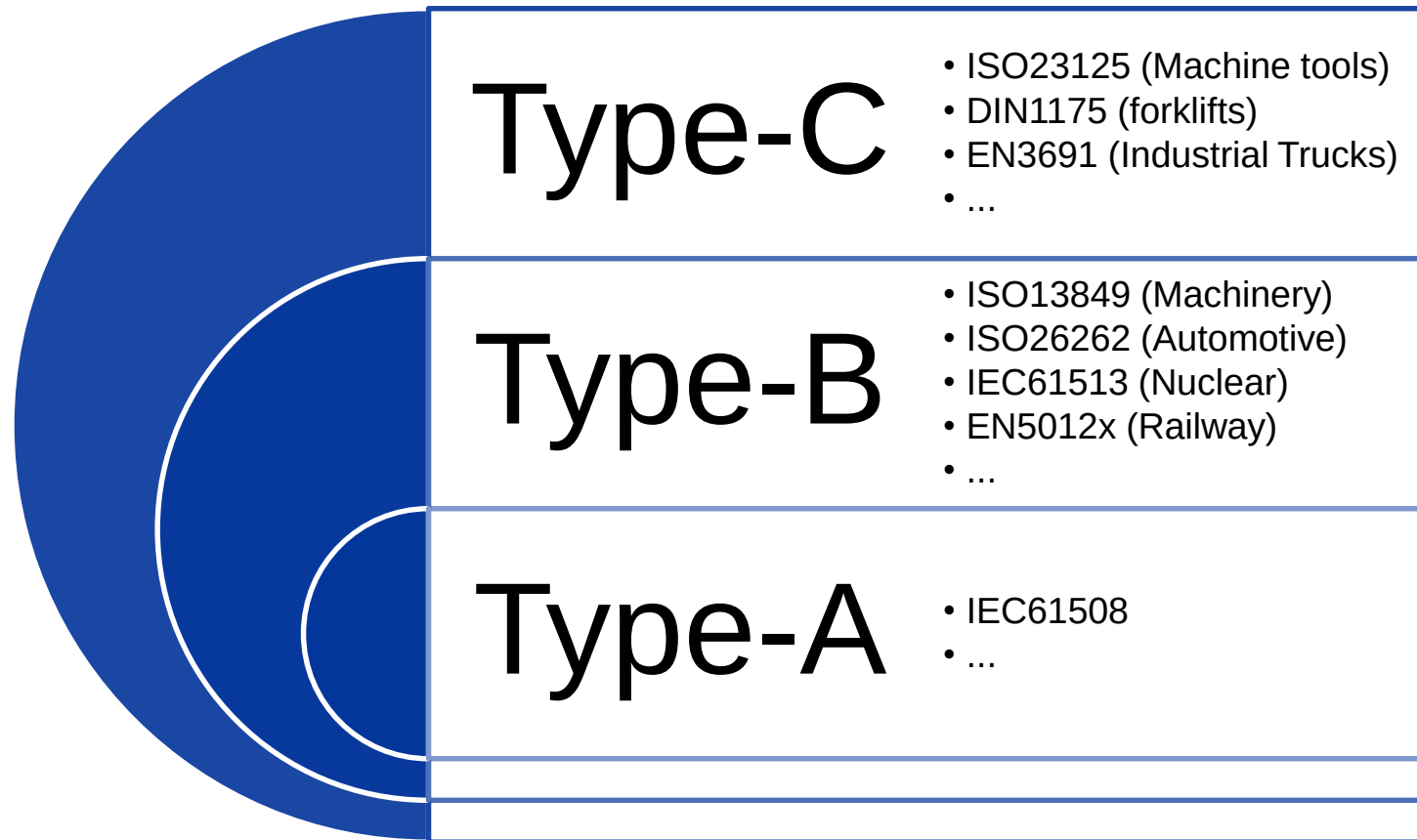
A small selection

- EN ISO 13849: Safety of Machines – focus machinery control systems
- EN/IEC 61508: Functional Safety of Electrical/electronic/programmable electronic safety-related systems
- EN/IEC 61511: Functional safety - Safety instrumented systems for the process industry sector
- EN/IEC 62061: Safety of machinery: Functional safety of electrical, electronic and programmable electronic control systems
- EN/IEC 50128: Railway applications - Communication, Signal Data Processing Systems - Software Railway Control Systems
- ISO 26262: Road vehicles – Functional safety

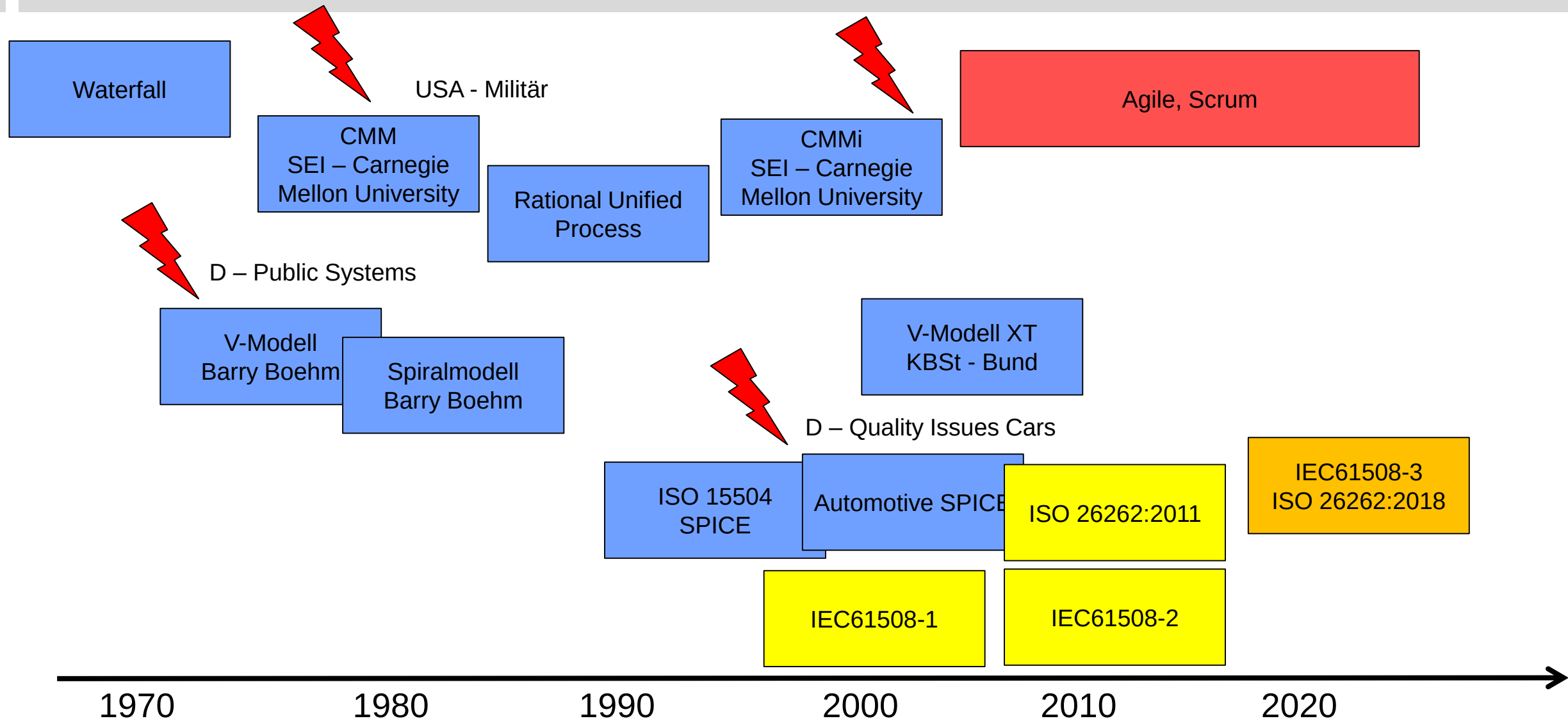
V.T.E		ISO standards by standard number	[hide]
		List of ISO standards / ISO romanizations / IEC standards	
1-9999	1	2 · 3 · 4 · 5 · 6 · 7 · 9 · 16 · 17 · 31 (0 - 1 · -2 · -3 · -4 · -5 · -6 · -7 · -8 · -9 · -10 · -11 · -12 · -13) · 128 · 216 · 217 · 226 · 228 · 233 · 259 · 269 · 302 · 306 · 361 · 428 · 500 · 518 · 519 · 639 (-1 · -2 · -3 · -5 · -6) · 646 · 657 · 668 · 690 · 704 · 732 · 764 · 838 · 843 · 860 · 898 · 965 · 999 · 1000 · 1004 · 1007 · 1073-1 · 1155 · 1413 · 1538 · 1629 · 1745 · 1989 · 2014 · 2015 · 2022 · 2033 · 2047 · 2108 · 2145 · 2146 · 2240 · 2281 · 2533 · 2709 · 2711 · 2720 · 2788 · 2848 · 2852 · 3029 · 3103 · 3166 (-1 · -2 · -3) · 3297 · 3307 · 3601 · 3602 · 3864 · 3901 · 3950 · 3977 · 4031 · 4157 · 4165 · 4217 · 4909 · 5218 · 5426 · 5427 · 5428 · 5725 · 5775 · 5776 · 5800 · 5807 · 5964 · 6166 · 6344 · 6346 · 6385 · 6425 · 6429 · 6438 · 6523 · 6709 · 6943 · 7001 · 7002 · 7010 · 7027 · 7064 · 7098 · 7185 · 7200 · 7498 (-1) · 7637 · 7736 · 7810 · 7811 · 7812 · 7813 · 7816 · 7942 · 8000 · 8093 · 8178 · 8217 · 8373 · 8501-1 · 8571 · 8583 · 8601 · 8613 · 8632 · 8651 · 8652 · 8691 · 8805/8806 · 8807 · 8820-5 · 8859 (-1 · -2 · -3 · -4 · -5 · -6 · -7 · -8 · -8-1 · -9 · -10 · -11 · -12 · -13 · -14 · -15 · -16) · 8879 · 9000/9001 · 9036 · 9075 · 9126 · 9141 · 9227 · 9241 · 9293 · 9314 · 9362 · 9407 · 9506 · 9529 · 9564 · 9592/9593 · 9594 · 9660 · 9797-1 · 9897 · 9899 · 9945 · 9984 · 9985 · 9995	
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20000+	20000	20022 · 20121 · 20400 · 21000 · 21047 · 21500 · 21827:2002 · 22000 · 22300 · 23270 · 23271 · 23360 · 24517 · 24613 · 24617 · 24707 · 25178 · 25964 · 26000 · 26262 · 26300 · 26324 · 27000 series · 27000 · 27001 · 27002 · 27006 · 27729 · 28000 · 29110 · 29148 · 29199-2 · 29500 · 30170 · 31000 · 32000 · 37001 · 38500 · 40500 · 42010 · 45001 · 50001 · 55000 · 80000 (-1)	
		Category	

Source: Wikipedia

Hierarchy of Safety Standards

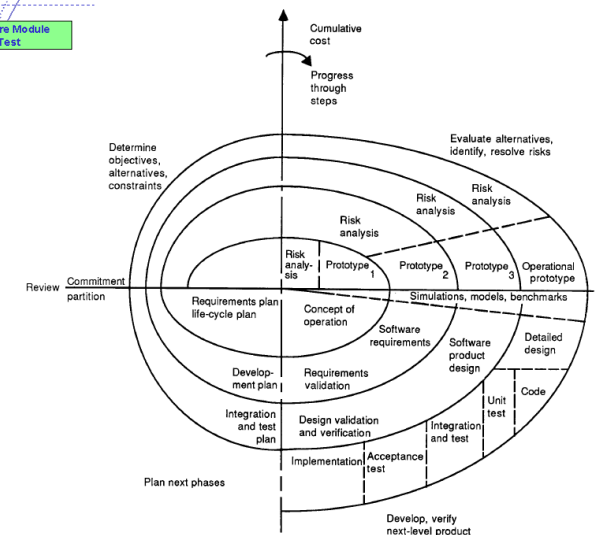
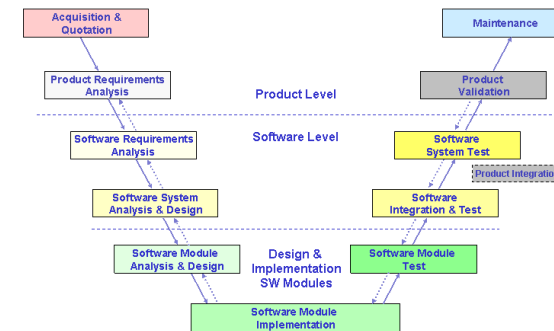
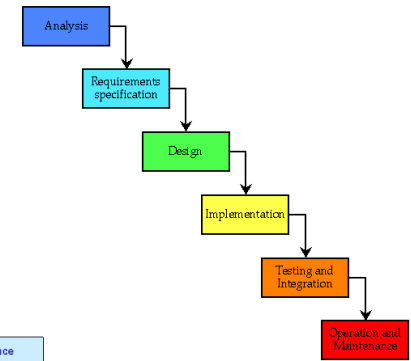


History of Processes and Standards



Phase 1 : Waterfall, V-Modell, Spiral (1956 – 1995)

- Waterfall – First very simple, non-iterative attempt to structure a software project into phases
- V-Model – Extending the waterfall, putting a stronger emphasis on testing
- Spiral – Realization, big bang does not work



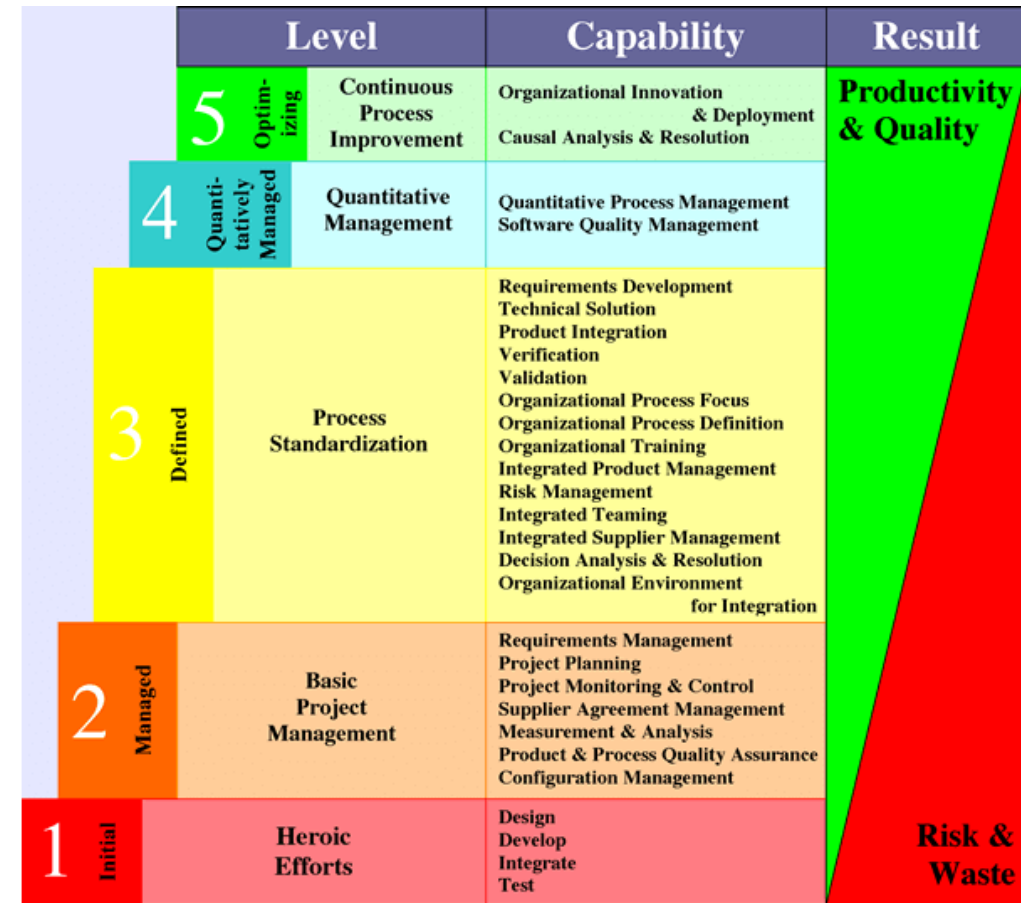
Phase 2 – CMM, CMMi, Spice (1995 – 2005)

A phase model is not enough, we need methods, roles, document templates

Process Dimension of ISO/IEC TR 15504

PRIMARY LIFE CYCLE PROCESSES		SUPPORTING LIFE CYCLE PROC.	
CUS.1 Acquisition	ENG.1 Development	SUP.1 Documentation	
CUS.1.1 Acquisition Preparation	ENG.1.1 System requirements analysis and design	SUP.2 Configuration management	
CUS.1.2 Supplier selection	ENG.1.2 Software requirements analysis	SUP.3 Quality Assurance	
CUS.1.3 Supplier monitoring	ENG.1.3 Software design	SUP.4 Verification	
CUS.1.4 Customer acceptance	ENG.1.4 Software construction	SUP.5 Validation	
CUS.2 Supply	ENG.1.5 Software integration	SUP.6 Joint Review	
CUS.3 Requirements elicitation	ENG.1.6 Software testing	SUP.7 Audit	
CUS.4 Operation	ENG.1.7 System integration and testing	SUP.8 Problem Resolution	
CUS.4.1 Operational use	ENG.2 System and software maintenance		
CUS.4.2 Customer support			
ORGANISATIONAL LIFE CYCLE PROCESSES			
MAN.1 Management	ORG.1 Organisational alignment	ORG.3 Human resource management	
MAN.2 Project Management	ORG.2 Improvement process	ORG.4 Infrastructure	
MAN.3 Quality Management	ORG.2.1 Process establishment	ORG.5 Measurement	
MAN.4 Risk Management	ORG.2.2 Process assessment	ORG.6 Reuse	
	ORG.2.3 Process improvement		

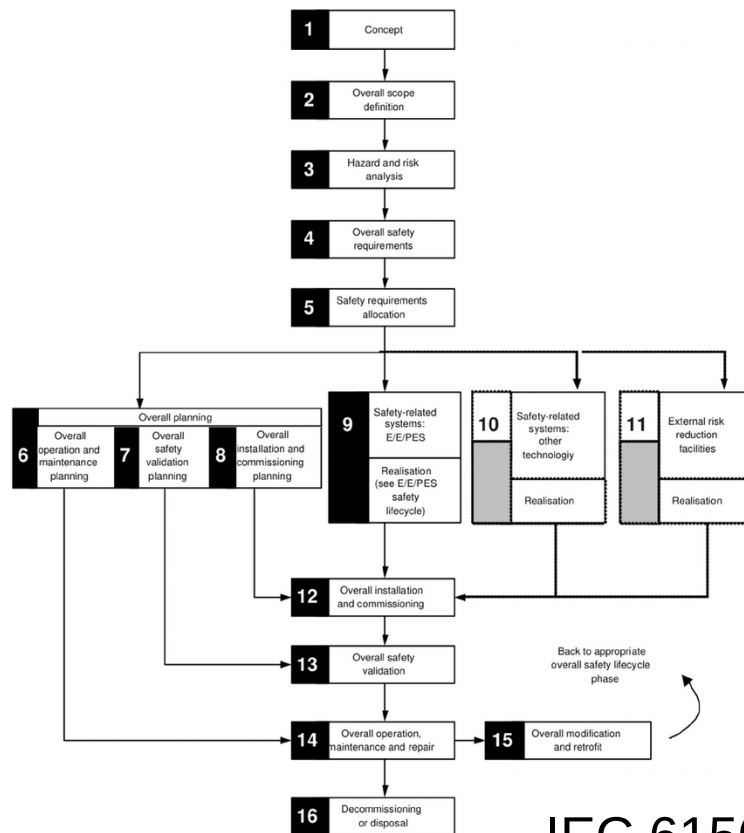
CMMi



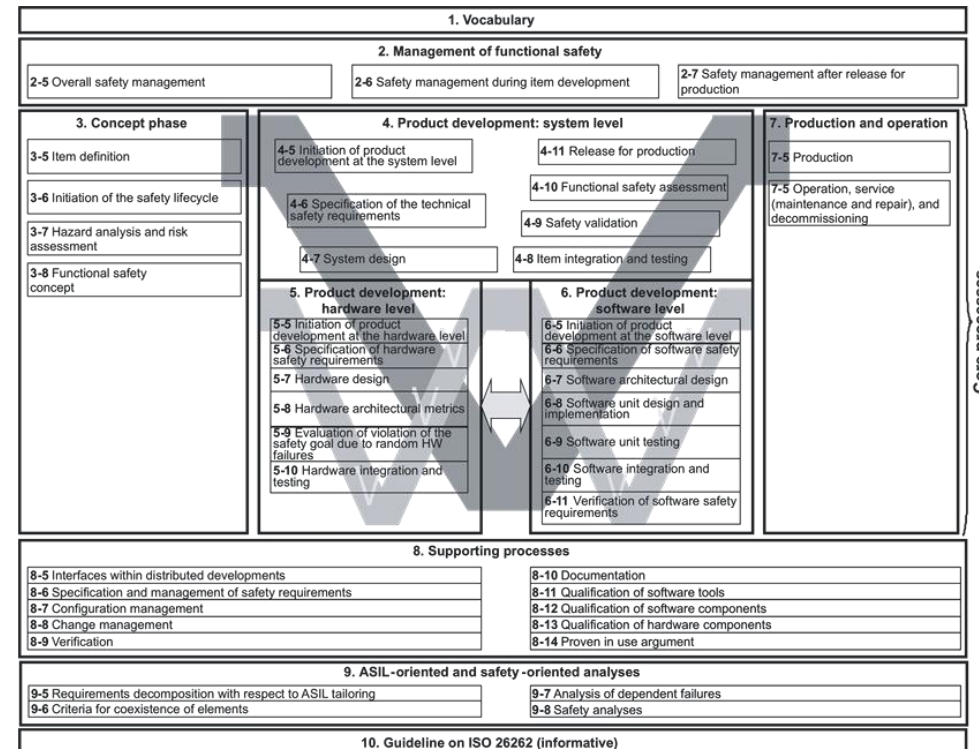
Source: www.researchgate.net

Phase 3 – Safety Standards (2000 ongoing)

Automation and Embedded Systems became more complex and safety critical.



IEC 61508



ISO 26262

Structure of the IEC61508 versus ISO26262

Part	Content
1	General requirements
2	Requirements for electrical/electronic/programmable electronic safety-related systems
3	Software requirements
4	Definitions and abbreviations
5	Examples of methods for the determination of safety integrity levels
6	Guidelines on the application of IEC 61508-2 and IEC 61508-3

New in v2018

Part	Content
1	Vocabulary
2	Management of Functional Safety
3	Concept Phase
4	Product Development at the System Level
5	Product Development at the Hardware Level
6	Product Development at the Software Level
7	Production and Operation
8	Supporting Processes
9	Automotive Safety Integrity Level (ASIL)-oriented and safety-oriented analyses
10	Guideline on ISO 26262
11	Guidelines on application of ISO 26262 to semiconductors
12	Adaptation of ISO 26262 for motorcycles

61508 Scope and Structure

Scope

- This International Standard covers those aspects to be considered when electrical / electronic / programmable electronic (E/E/PE) systems are used to carry out safety functions.
- A major objective of this standard is to facilitate the development of product and application sector international standards by the technical committees responsible for the product or application sector.
- ...
- A second objective of this standard is to enable the development of E/E/PE safety-related systems where product or application sector international standards do not exist.

Source for this and all following slides: IEC 61508

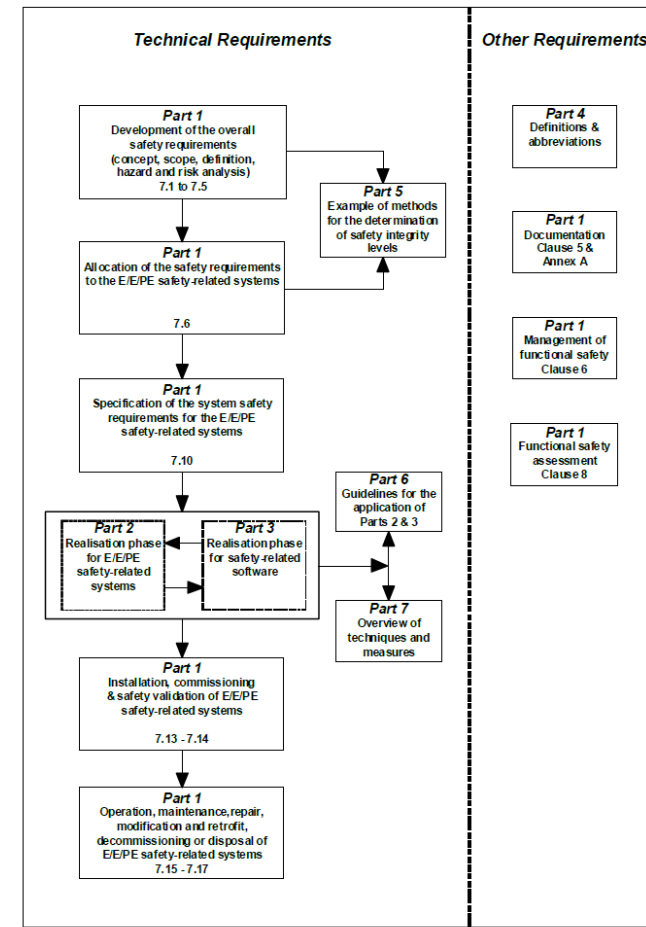


Figure 1 – Overall framework of the IEC 61508 series

61508 Part 1 – General Requirements

4.1 To conform to this standard it shall be demonstrated that all the relevant requirements have been satisfied to the required criteria specified (for example safety integrity level) and therefore, for each clause or subclause, all the objectives have been met.

Chapter 5 - Documentation

- Documentation of design and verification principles
- Documentation of the development process

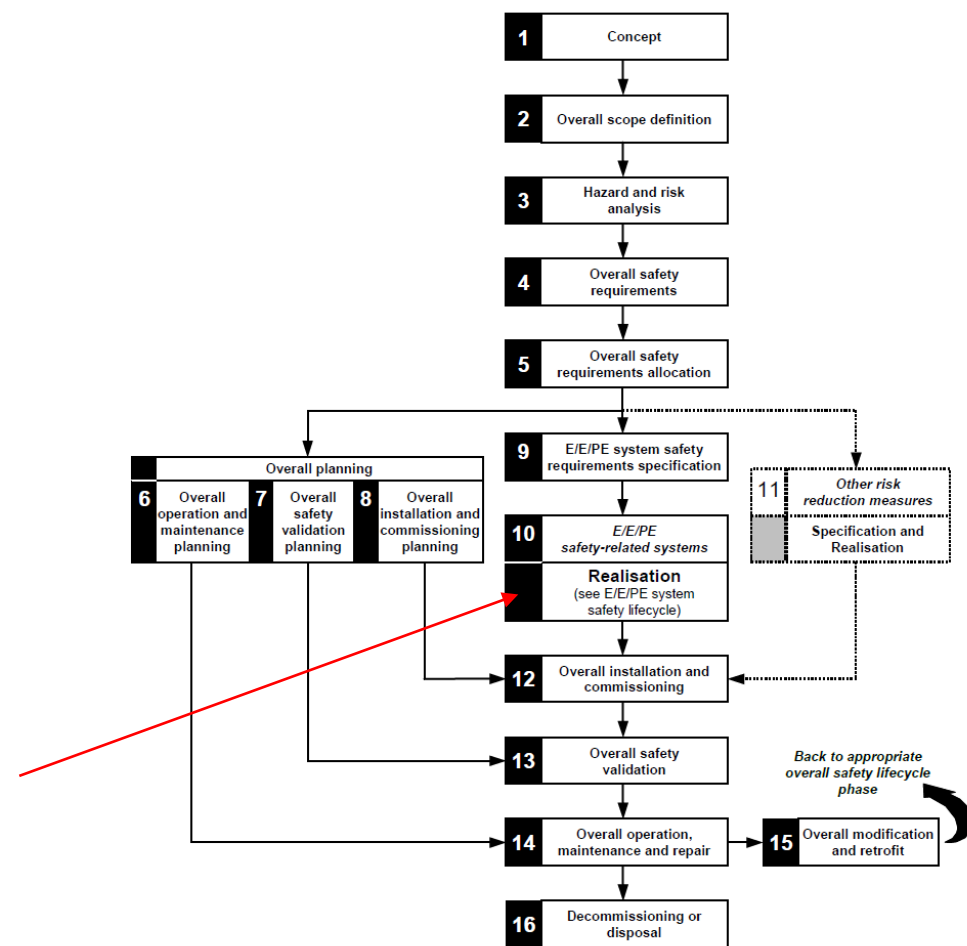
5.2.6 The documentation shall:

- be accurate and concise;
- be easy to understand by those persons having to make use of it;
- suit the purpose for which it is intended;
- be accessible and maintainable.

Chapter 6 - Management of Functional Safety

- Definition of responsibilities
 - Independent safety responsibility
 - Definition of interfaces (organizational)
 - Execution of self assessments
- and processes
 - Hazard and risk analysis
 - Verification Activities
 - Validation Activities
 - Configuration Management
 - Incident Reporting and Analysis

61508 Chapter 7 – Overall Safety Lifecycle Requirements



Details



Table 1 – Overall safety lifecycle – overview

Safety lifecycle phase							
Figure 2 box number	Title	Objectives	Scope	Requirements subclause	Inputs	Outputs	
1	Concept	7.2.1: To develop a level of understanding of the EUC and its environment (physical, legislative etc.) sufficient to enable the other safety lifecycle activities to be satisfactorily carried out.	EUC and its environment (physical, legislative etc.)	7.2.2	All relevant information necessary to meet the requirements of the subclause.	Information concerning the EUC, its environment and hazards.	
2	Overall scope definition	7.3.1: To determine the boundary of the EUC and the EUC control system; To specify the scope of the hazard and risk analysis (for example process hazards, environmental hazards, etc.)	EUC and its environment.	7.3.2	Information concerning the EUC, its environment and hazards.	Defined scope of the hazard and risk analysis.	
3	Hazard and risk analysis	7.4.1: To determine the hazards, hazardous events and hazardous situations relating	The scope will be dependent upon the phase reached in the	7.4.2	Defined scope of the hazard and risk analysis.	Description of, and information relating to, the	

61508 Chapter 7 – Overall Safety Lifecycle Requirements

Hazard and Risk Identification

Hazard and risk analysis

- The first objective of the requirements of this subclause is to determine the hazards, hazardous events and hazardous situations relating to the EUC and the EUC control system (in all modes of operation) for **all reasonably foreseeable circumstances, including fault conditions and reasonably foreseeable misuse**

The hazard and risk analysis shall consider the following:

- each determined hazardous event and the components that contribute to it;
- the consequences and likelihood of the event sequences with which each hazardous event is associated;
- the tolerable risk for each hazardous event;
- the measures taken to reduce or remove hazards and risks;
- the assumptions made during the analysis of the risks, including the estimated demand rates and equipment failure rates; any credit taken for operational constraints or human intervention shall be detailed.

EUC = Equipment under Control

Hazard and Risk Analysis

Top Down

Can a hazardous situation occur?

How can it be triggered?

How do we evaluate this hazardous event?

Risk minimization
(Safety Requirements)

Hazard analysis

“Which damage can be achieved by the EUC?”

Hazardous event 1

Hazardous event n

Risk assessment

- Severity
- Probability
- Controllability
- ...

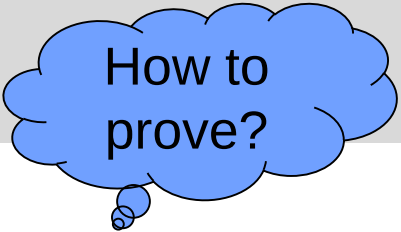
Safety Function

- Design
- Realization
- Verification
- Assessment

SIL_{req}

SIL_{ach}

SIL required versus SIL achieved (See 61508 part 5)



E.6 Risk graph example

An example of a risk graph implementation based on the example data in Table E.1 below is shown in Figure E.2. Use of the risk parameters C, F, and P lead to one of eight outputs. Each one of these outputs is mapped onto one of three scales (W1, W2 and W3). Each point on these scales (a, b, c, d, e, f, g and h) is an indication of the necessary risk reduction that has to be met by the safety-related system.

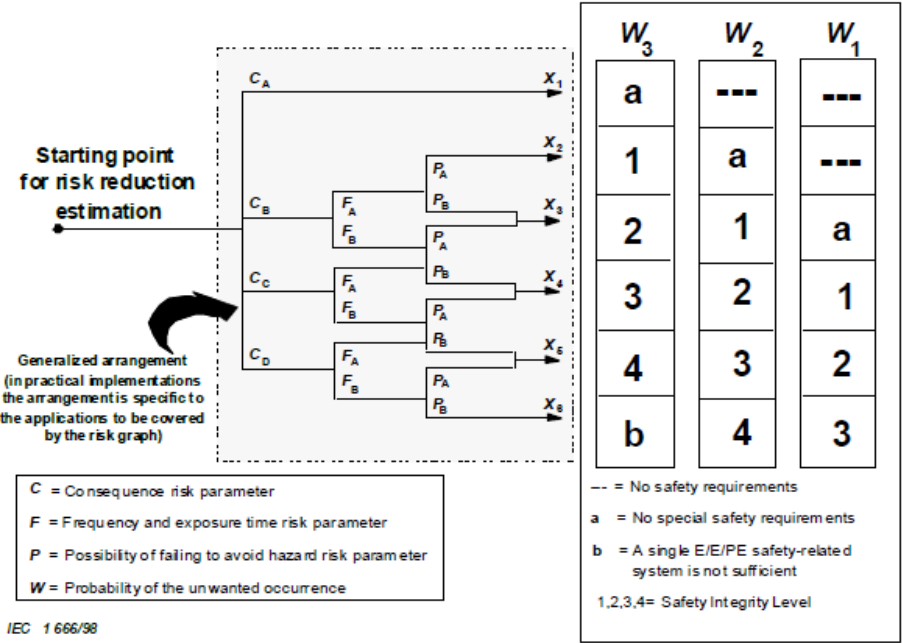


Table 2 – Safety integrity levels – target failure measures for a safety function operating in low demand mode of operation

Safety integrity level (SIL)	Average probability of a dangerous failure on demand of the safety function (PFD _{avg})
4	$\geq 10^{-5}$ to $< 10^{-4}$
3	$\geq 10^{-4}$ to $< 10^{-3}$
2	$\geq 10^{-3}$ to $< 10^{-2}$
1	$\geq 10^{-2}$ to $< 10^{-1}$

e.g. Protection Function

Table 3 – Safety integrity levels – target failure measures for a safety function operating in high demand mode of operation or continuous mode of operation

Safety integrity level (SIL)	Average frequency of a dangerous failure of the safety function [h ⁻¹] (PFH)
4	$\geq 10^{-9}$ to $< 10^{-8}$
3	$\geq 10^{-8}$ to $< 10^{-7}$
2	$\geq 10^{-7}$ to $< 10^{-6}$
1	$\geq 10^{-6}$ to $< 10^{-5}$

e.g. Safe System Function

criticality

reliability

Difference between SIL required and SIL achieved → additional protection required!

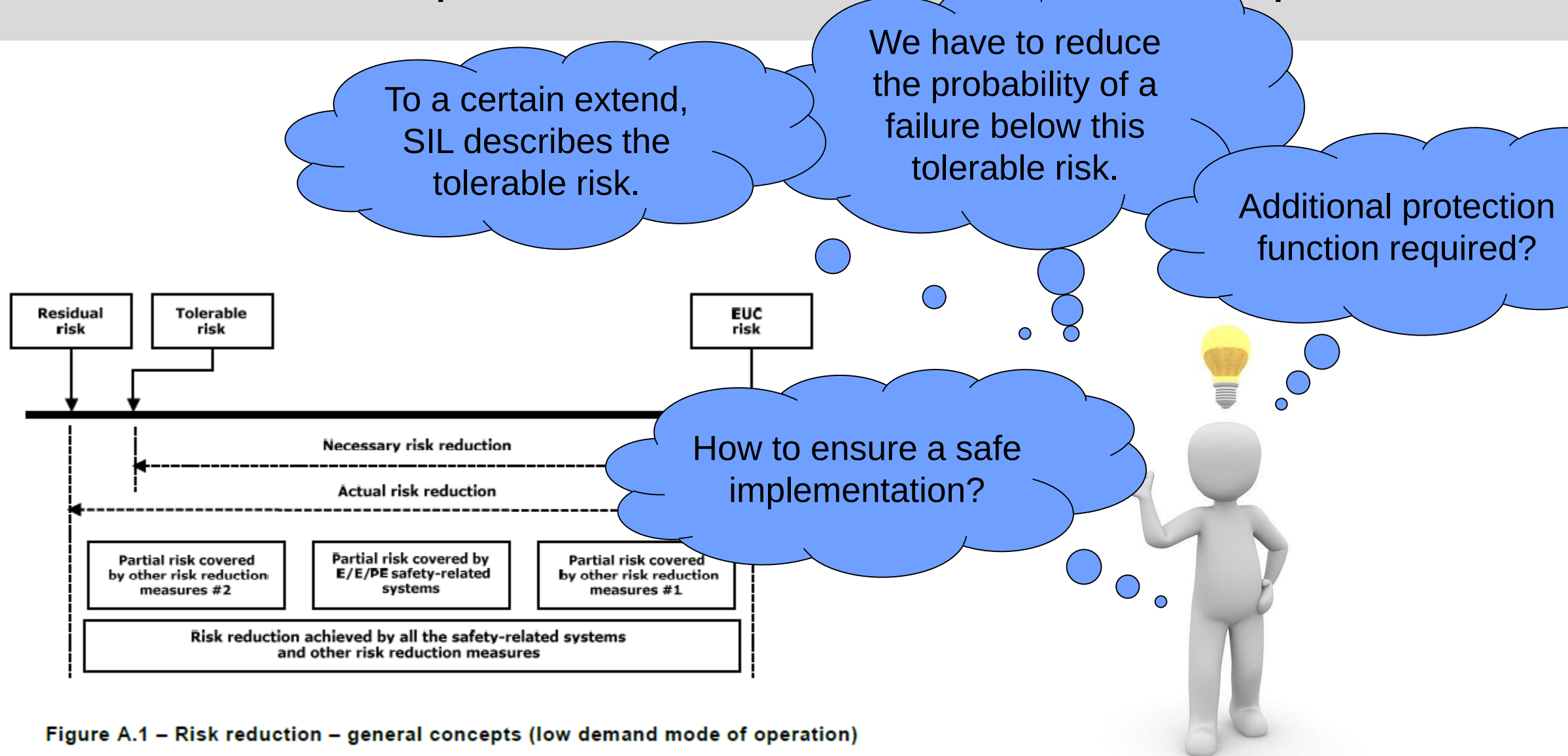
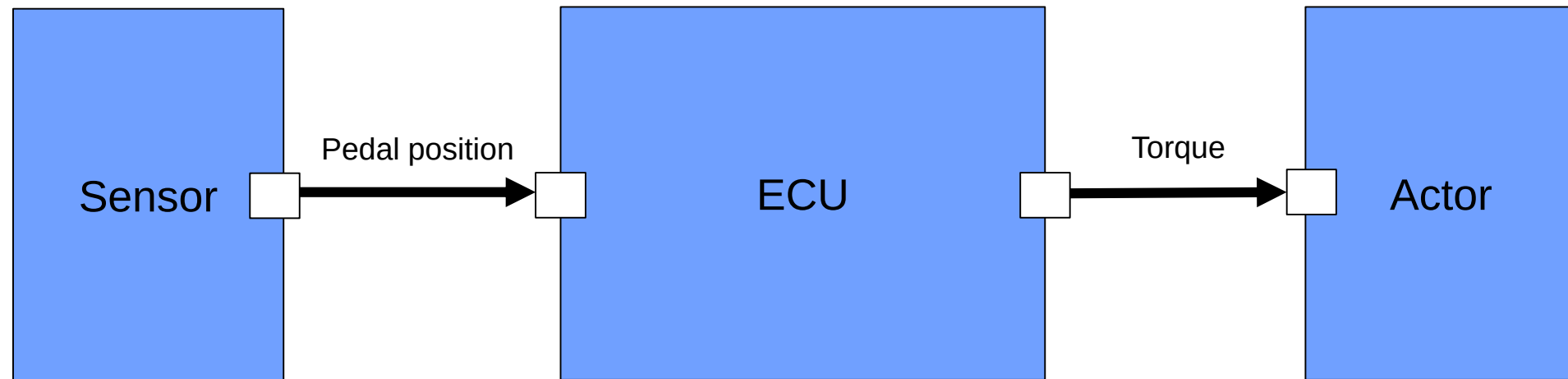


Figure A.1 – Risk reduction – general concepts (low demand mode of operation)

Example: Electronic Throttle Control

- Can a hazard happen?
- Which events may lead to this hazard?
- How do we evaluate the risks?



Example Electronic Throttle Control

Hazard

- H1 - The car is involved in an accident due to wrong speed
- H2 – The car is hurting a pedestrian due to unintended start of a motion

Hazardous Events (subset)

- E1 – Ignition: Unintended Engine Start
- E2 – Ignition: Engine is not correctly turned off
- E3 – Targetspeed: Car does not speed up when driver presses gas pedal
- E4 – Targetspeed: Car does not slow down when releasing the gas pedal (but we still can brake)
- E5 - Targetspeed: Car is speeding up without driver interaction (we still can brake)

Risk assessment (subset)

- E3 – Targetspeed: Car is speeding up without driver interaction (we still can brake)
 - Severity: High
 - Probability: Medium
 - Controlability: Medium
- Technical reason: Potentiometer is providing wrong value

We want to achieve SIL 3
(just as an example)
PFH 10^{-8} ... 10^{-7}

Example Electronic Throttle Control

Acc. table 1: KFD2-PT2-Ex1-* Potentiometer converter 1001 structure

Parameters acc. to IEC61508	Variables
Device type	A
Demand mode	Low demand mode or High demand mode
Safety Function	Potentiometer converter
HFT	0
SIL	2
λ_s	0 FIT
λ_{dd}	0 FIT
λ_{du}	86.1 FIT
$\lambda_{fail\ high}^1$	6.35 FIT
$\lambda_{fail\ low}^1$	86.5 FIT
$\lambda_{no\ effect}$	218 FIT
$\lambda_{total\ (Safety\ function)}$	397 FIT
SFF	78.31%
MTBF ²	265 years
PFH	$8.62 \cdot 10^{-8} \text{ 1/h}$
PFD _{avg} for $T_1 = 1 \text{ year}$	$3.77 \cdot 10^{-4}$
T _{proof,max}	2.5 years

¹ "Fail high" and "fail low" failures are considered as dangerous detected failures.
² acc. To SN29500. This value includes failures which are not part of the safety function / MTTR = 8h

Table 3 – Safety integrity levels – target failure measures for a safety function operating in high demand mode of operation or continuous mode of operation

Safety integrity level (SIL)	Average frequency of a dangerous failure of the safety function [h ⁻¹] (PFH)
4	$\geq 10^{-9} \text{ to } < 10^{-8}$
3	$\geq 10^{-8} \text{ to } < 10^{-7}$
2	$\geq 10^{-7} \text{ to } < 10^{-6}$
1	$\geq 10^{-6} \text{ to } < 10^{-5}$

$SIL_{req} < SIL_{ach}$ (ok for the Poti)

➔ But how about additional errors?

Introduce an additional safety function (e.g. a redundant second sensor channel.)

CONFIDENTIAL acc. to ISO 16016	Only valid as long as released in EDM or with a valid production documentation!	scale: 1:1	date: 2010-Nov-24
f PEPPERL+FUCHS	FIMEDA – Report	respons. DP.HSU	FS-0058PF-20A
Mannheim	KFD2-PT2-Ex1-*	approved	
template: FTM-0027_1		norm	sheet 4 of 10

61508 Chapter 7 – Overall Safety Lifecycle Requirements

Overall Safety Requirements

7.5.2.1 A set of all necessary overall safety functions shall be developed based on the hazardous events derived from the hazard and risk analysis. This shall constitute the specification for the overall safety functions requirements.

The data provided by the pedal position must be reliable.

We need to detect short circuit and open circuit hardware errors.

We need to detect “sticky pins”

We need to detect problems with the ADC



Design and Implementation

- The hazard and risk analysis tells us which SIL level we require
- Now we only have to figure out which SIL our system actually has – and improve it when required

Hardware

- Reliability of components
- Hardware design
 - Redundancy
 - Common cause failure analysis
 - Freedom from interference
- Verification and validation / FMEDA / FMA
- ...

Part 2

Software

- Reliability of development process
- Software design
 - Diagnostic coverage
 - Freedom from interference
- Validation and Verification
- ...

Part 3

61508 Chapter 7 – Overall Safety Lifecycle Requirements

The right side of the V

Overall operation and maintenance planning

- Safety audit documentation
- Safety constraints of the EUC
- Maintenance requirements

Overall safety validation planning

- Validation plan
 - Operational modes to be validated
 - Validation methods to be applied
-
- Overall installation and commissioning planning
 - Installation procedure
 - Procedures for problem management

61508 Chapter 8 – Functional Safety Assessment

The goal of the assessment is to check if the process followed and actions taken are in line with the requirements of the IEC61508 Standard.

8.2.1 One or more persons shall be appointed to carry out one or more functional safety assessments in order to arrive at a judgement on the adequacy of:

- the functional safety achieved by the E/E/PE safety-related systems, within their particular environment, in respect to the relevant clauses of this standard;
- the compliance to the relevant clauses of this standard, achieved in the case of elements/subsystems.

8.2.2 Those carrying out a functional safety assessment shall have access to all persons involved in any overall, E/E/PE system or software safety lifecycle activity and all relevant information and equipment (both hardware and software).

NOTE It is recognised that access to those persons who were previously involved in a safety lifecycle phase may not be achievable and in such a case reliance has necessarily to be placed on those persons currently having relevant responsibilities.

Who should conduct such an assessment?

Table 4 – Minimum levels of independence of those carrying out functional safety assessment (overall safety lifecycle phases 1 to 8 and 12 to 16 inclusive (see Figure 2))

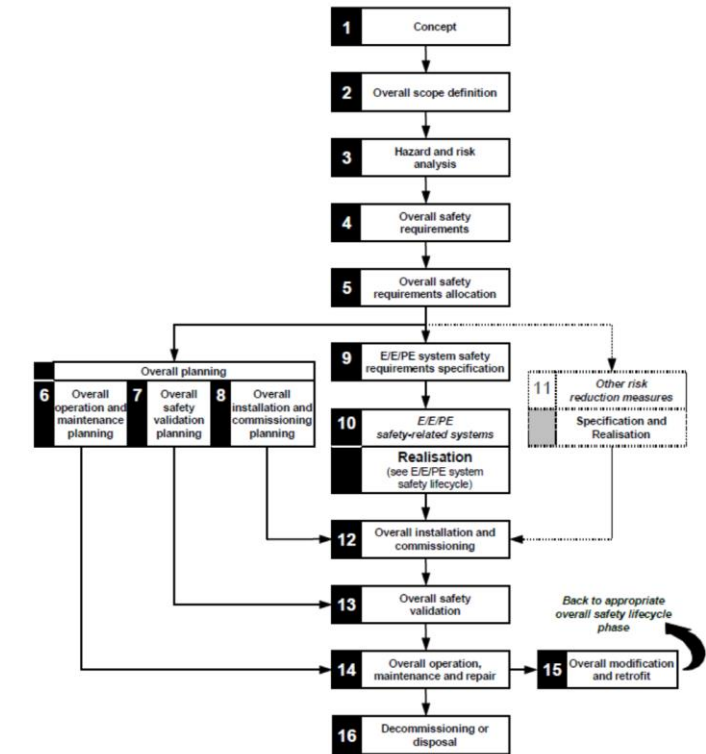
Minimum level of independence	Consequence (see 8.2.17)			
	A	B	C	D
Independent person	X	X1	Y	Y
Independent department		X2	X1	Y
Independent organization			X2	X

NOTE See 8.2.15, 8.2.16 and 8.2.17 for details on interpreting this table.

Table 5 – Minimum levels of independence of those carrying out functional safety assessment (overall safety lifecycle phases 9 and 10, including all phases of E/E/PE system and software safety lifecycles (see Figures 2, 3 and 4))

Minimum level of independence	Safety integrity level/Systematic capability			
	1	2	3	4
Independent person	X	X1	Y	Y
Independent department		X2	X1	Y
Independent organization			X2	X

NOTE See 8.2.15, 8.2.16 and 8.2.18 for details on interpreting this table.

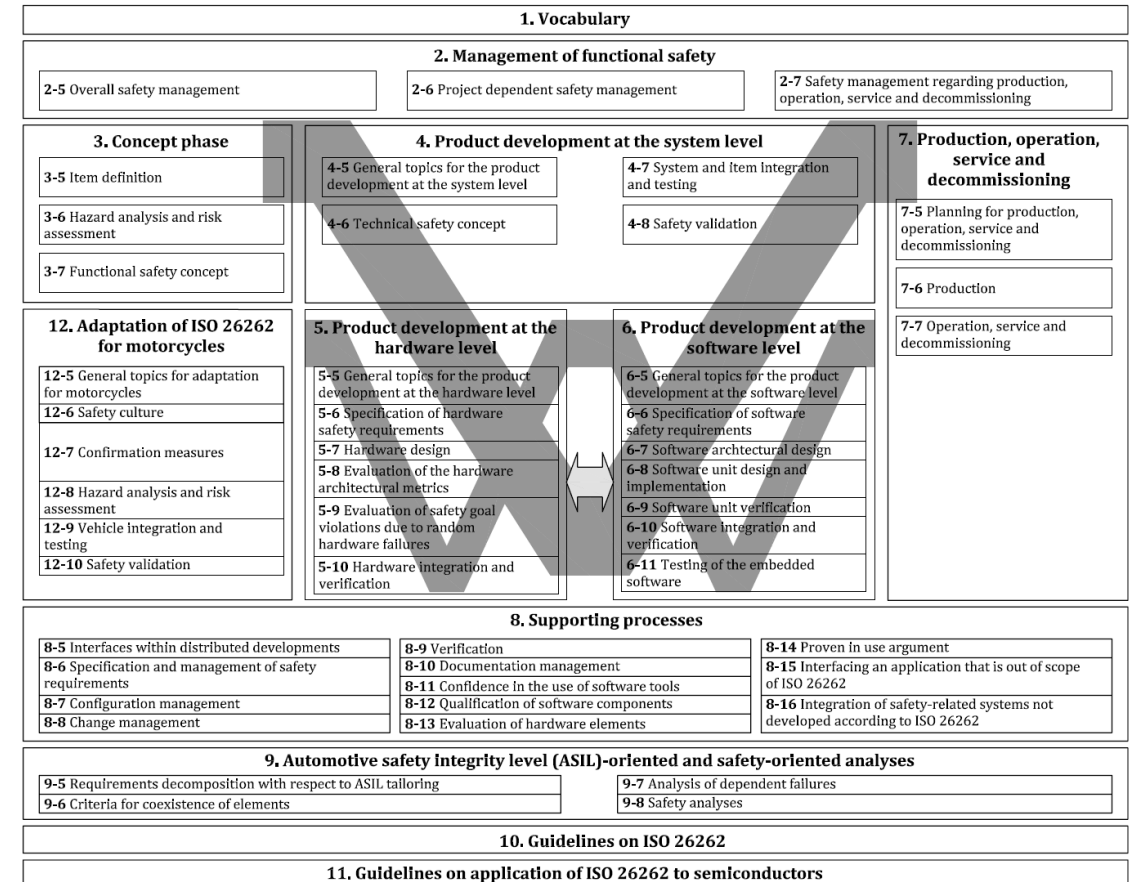


8.2.17 In the context of Table 4, the consequence values for the specified level of independence are:

- Consequence A: minor injury (for example temporary loss of function);
- Consequence B: serious permanent injury to one or more persons, death to one person;
- Consequence C: death to several people;
- Consequence D: very many people killed.

61508 versus 26262

- Type B standard, only valid for passenger cars (and motorcycles v2018)
- Based on 61508 (especially general lifecycle concepts comparable)
- More specific in terms of risk assessment – focusing on typical driving situations
- Method recommendations slightly more state of the art and specific



Source: All quotes from here on based on ISO 26262:2018

ISO26262 Hazard and Risk Analysis

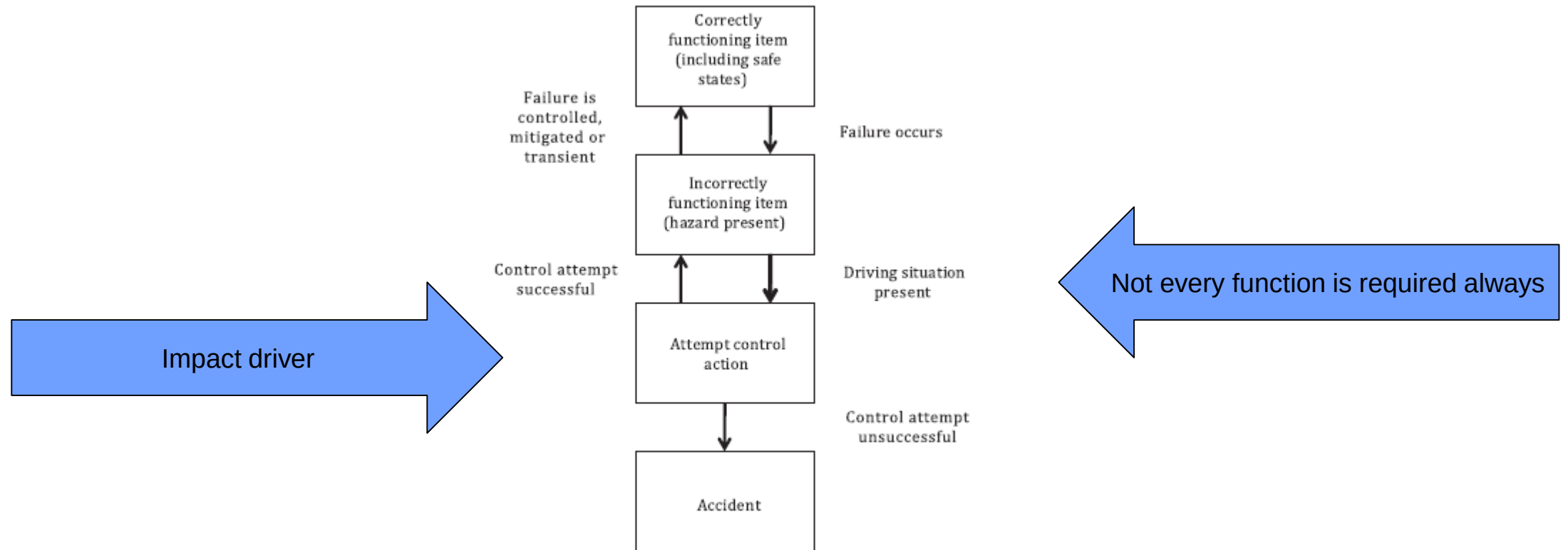


Figure 2 — State machine model of automotive risk

ISO26262 Hazard and Risk Analysis

- ASIL A...D (highest) + QM (Quality Management)

Table 4 — ASIL determination

Severity class	Exposure class	Controllability class		
		C1	C2	C3
S1	E1	QM	QM	QM
	E2	QM	QM	QM
	E3	QM	QM	A
	E4	QM	A	B
S2	E1	QM	QM	QM
	E2	QM	QM	A
	E3	QM	A	B
	E4	A	B	C
S3	E1	QM	QM	A ^a
	E2	QM	A	B
	E3	A	B	C
	E4	B	C	D

^a See [6.4.3.11](#).

	Class			
	S0	S1	S2	S3
Description	No injuries	Light and moderate injuries	Severe and life-threatening injuries (survival probable)	Life-threatening injuries (survival uncertain), fatal injuries

Table 2 — Classes of probability of exposure regarding operational situations

	Class				
	E0	E1	E2	E3	E4
Description	Incredible	Very low probability	Low probability	Medium probability	High probability

	Class			
	C0	C1	C2	C3
Description	Controllable in general	Simply controllable	Normally controllable	Difficult to control or uncontrollable

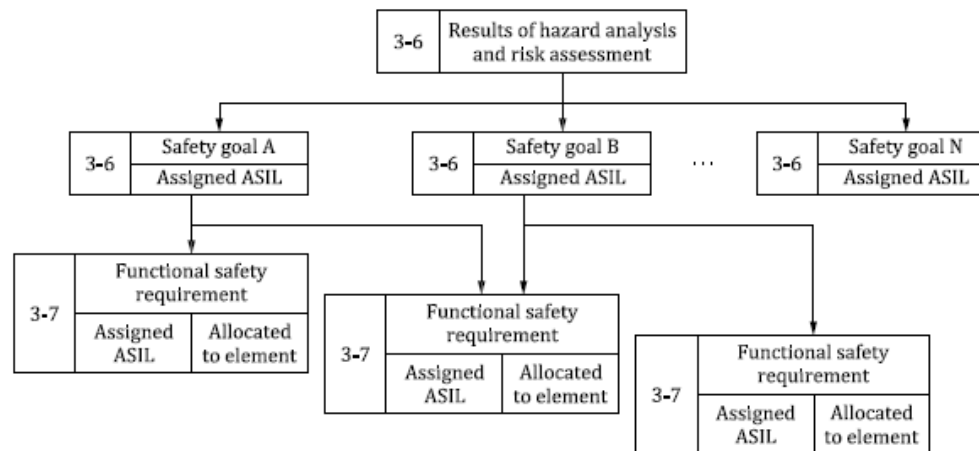
ISO26262 - Example Hazard Analysis

Table 1 — Safety goals resulting from the same hazard in different situations

Failure mode	Hazard	Specific situation	Hazardous event	Possible consequences	ASIL	Safety goal	Safe state
Unintended parking brake activation	Unexpected deceleration	High speed OR taking a bend OR low friction surface	Unexpected deceleration at high speed OR taking a bend OR low friction surface	Loss of vehicle stability	<i>Higher ASIL</i>	Avoid activating the parking function without the driver's request when the vehicle is moving	EPB disabled
Unintended parking brake activation	Unexpected deceleration	Medium-low speed AND high friction surface	Unexpected deceleration at medium-low speed AND high friction surface	Rear end collision with the following vehicle	<i>Lower ASIL</i>	Avoid activating the parking function without the driver's request when the vehicle is moving	EPB disabled

ISO26262 Safety Goals and Safety Requirements

Example:



- Safety Goal: The car speed shall be proportional to the gas pedal position
- Safety Requirement 1: Reliable Sensor, allocated to element Potentiometer
- Safety Requirement 2: The control function shall be called every 10ms, allocated to element OS / control flow monitoring
- ...

Wrapup

- IEC 61508 Part 1 (Overall Safety Lifecycle)
 - Safety Integrity Level as key term
 - Hazard and Risk Analysis
 - Comparison SIL_{req} and SIL_{ach}
- Comparison IEC61508 and ISO26262
 - More specific type B standard
 - Focussed on typical passenger car driving scenarios